

TELECOM ANALYTICS PROJECT

ChurnGuard

A Customer Risk Intelligence System
for Telecom Revenue Protection

An End-to-End Analytics Solution Using Python, SQL & Power BI

Prepared by

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1. Executive Summary

This report presents the findings of a comprehensive customer risk and revenue intelligence analysis conducted on a telecommunications customer dataset. The project was executed using a full end-to-end analytics pipeline — Python for data processing and risk modelling, SQL for data validation and querying, and Power BI for executive dashboard delivery.

The objective was to identify customers at risk of churning, quantify the associated revenue exposure, and deliver a structured, data-driven framework enabling proactive retention strategies that prevent revenue loss before it becomes permanent.

7,000	GH¢244K	39.71	140
Total Customers Analysed	Total Revenue at Risk	Avg Risk Score (out of 250)	Urgent Action Customers

Key Findings at a Glance

- Approximately 7,000 customers analysed across three risk segments: Monitor, At Risk, and Urgent Action.
- An estimated GH¢244,330 in total revenue is at risk of being lost to churn.
- The Urgent Action segment — just 2% of the base — carries the highest revenue risk intensity per customer.
- 2,442 customers exhibit high spending volatility, a strong early churn indicator.
- A targeted retention framework can protect the majority of at-risk revenue through three distinct strategies.

2. Business Problem

Telecommunication companies operate in an increasingly competitive environment where customer churn is a persistent and costly challenge. Factors driving churn include:

- Aggressive competitor pricing and promotional offers
- Customer dissatisfaction with service quality or network reliability
- Declining usage patterns indicating early-stage disengagement
- Absence of proactive retention strategies before customers decide to leave

The financial consequences of unchecked churn are significant:

- Loss of predictable recurring revenue from departing customers
- Increased customer acquisition costs to replace churned subscribers
- Erosion of Customer Lifetime Value (CLV) across the subscriber base

Core Business Challenge

- How can a telecom operator identify high-risk customers early — before they churn — and take targeted, cost-effective actions to protect revenue?
- ChurnGuard answers that question with data, not guesswork, by building a predictive risk intelligence system that flags customers before revenue is lost.

This challenge mirrors real-world scenarios faced by major telecom operators across Africa, including MTN Ghana, Telecel, and AirtelTigo, where early identification of at-risk subscribers enables significant revenue protection.

3. Data & Methodology

3.1 Data Source

The analysis is built on the IBM Telco Customer Churn dataset, enhanced through feature engineering to simulate real-world time-based spending patterns representative of actual telecom customer behaviour.

3.2 Data Processing — Python

- Data cleaning: type correction, null handling, and consistency validation
- Construction of a time-based customer spending dataset simulating monthly behaviour
- Feature engineering: rolling average spend, month-over-month spend change, decline flags
- Calculation of revenue at risk per customer based on spending trajectory
- Export of processed dataset to SQL for validation and aggregation

3.3 Risk Scoring Model

A composite risk_score was engineered for each customer using three behavioural dimensions:

Risk Dimension	Description	Weight
Spend Decline Frequency	How often the customer's monthly spend has decreased	High
Revenue Drop Magnitude	The size of the most recent revenue decline	High
Spending Behaviour Pattern	Overall volatility in the customer spending history	Medium

Based on the composite score, customers were segmented into three action tiers:

Segment	Risk Level	Score Range	Action Required
Monitor	Low	0 — 30	No immediate action — observe and track
At Risk	Moderate	31 — 80	Engagement campaign and usage incentives
Urgent Action	High	81 — 250	Direct call, retention offer, priority intervention

3.4 Power BI Dashboard Delivery

- Dashboard 1: Customer Risk & Revenue Intelligence — Executive overview

- Dashboard 2: Customer Risk Deep Dive & Targeting — Individual customer analysis
- Dashboard 3: Customer Behaviour & Risk Drivers — Spending patterns and volatility
- Dashboard 4: Review Impact & Retention Strategy — Action prioritisation framework

4. Dashboard Walkthrough

4.1 Dashboard 1: Customer Risk & Revenue Intelligence

The executive overview dashboard provides leadership with an instant high-level view of total revenue exposure, customer risk distribution, and segment breakdown. It is designed to answer one question in under 30 seconds: How much revenue are we at risk of losing, and where is it concentrated?

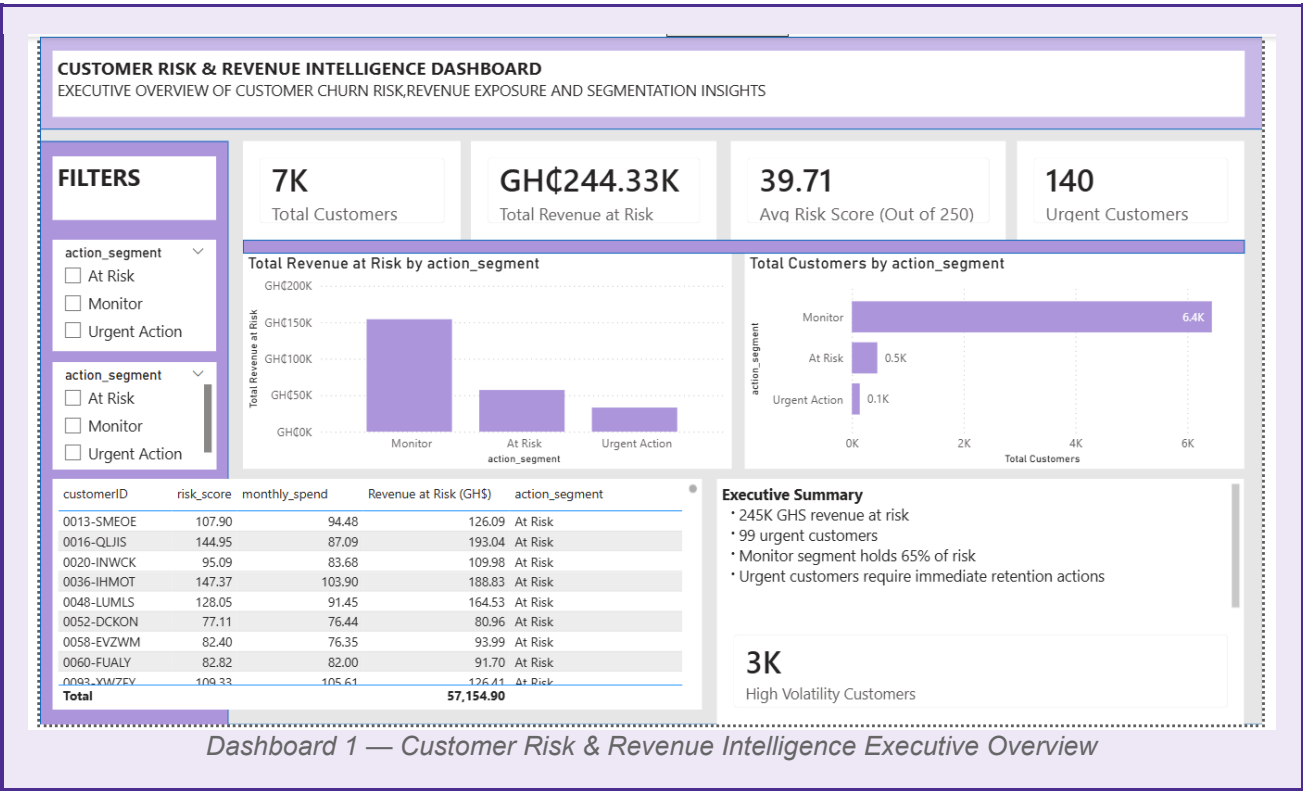


Figure 1: Executive Overview Dashboard — KPI cards, revenue by segment, and customer distribution

Key Metrics

Metric	Value	Interpretation
Total Customers	7,000	Full subscriber base analysed
Total Revenue at Risk	GH¢244,330	Potential revenue loss from churn
Average Risk Score	39.71 / 250	Moderate overall risk — tail segments require action
Urgent Customers	99	Require immediate retention intervention

Dashboard 1 Insight

- The Monitor segment holds the largest share of total revenue at risk (GH¢154,523) due to sheer volume.
- The Urgent Action segment's 140 customers represent the highest risk intensity per customer and require immediate priority.
- The donut chart confirms 92% of customers are in Monitor — a stable base, but the tail risk is significant.

4.2 Dashboard 2: Customer Risk Deep Dive & Targeting

The deep dive dashboard enables analysts and retention teams to drill into individual customer risk profiles. It surfaces the highest-risk customers by name, quantifies individual revenue exposure, and maps risk scores against monthly spending to identify targeting priorities.

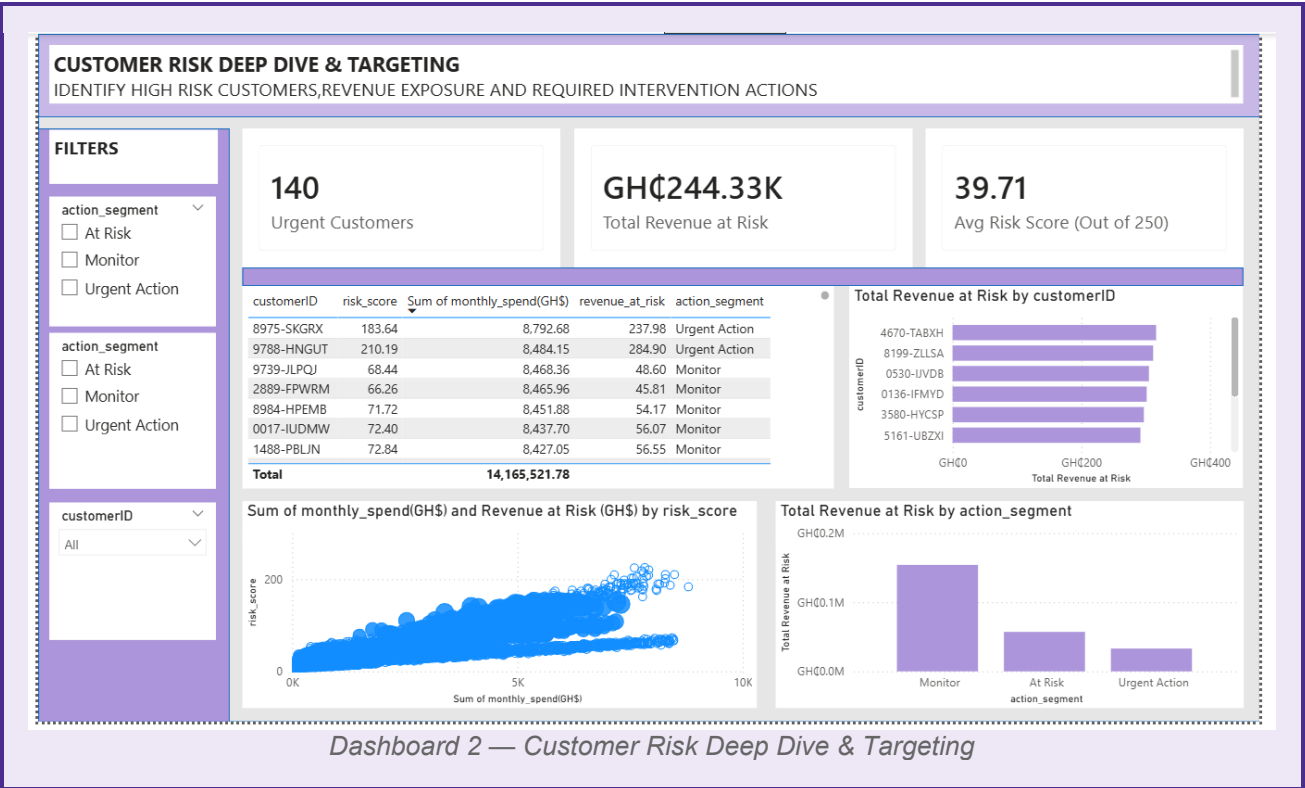


Figure 2: Deep Dive Dashboard — individual risk scores, revenue exposure, and scatter analysis

Top High-Risk Customers by Revenue at Risk

Rank	Customer ID	Risk Level	Segment	Action
1	9866-OCCKE	Highest	Urgent Action	Call immediately
2	2988-QRAJY	High	Urgent Action	Call immediately
3	5360-LJCNJ	High	Urgent Action	Call immediately
4	3655-SNQYZ	High	Urgent Action	Call immediately
5	1090-PYKCI	High	Urgent Action	Call immediately

Dashboard 2 Insight

- The scatter plot reveals: customers with higher monthly spend carry higher risk scores — high-value customers are most at risk.
- The revenue bar chart by Customer ID provides a ready-made priority call list for the retention

team.

- Filtering by Urgent Action narrows to 140 customers requiring contact within 72 hours.

4.3 Dashboard 3: Customer Behaviour & Risk Drivers

The behaviour dashboard examines the underlying drivers of churn risk — spending volatility, usage patterns, and how risk concentrates across segments. This dashboard answers: Why are customers becoming high risk, and which behaviours predict churn earliest?

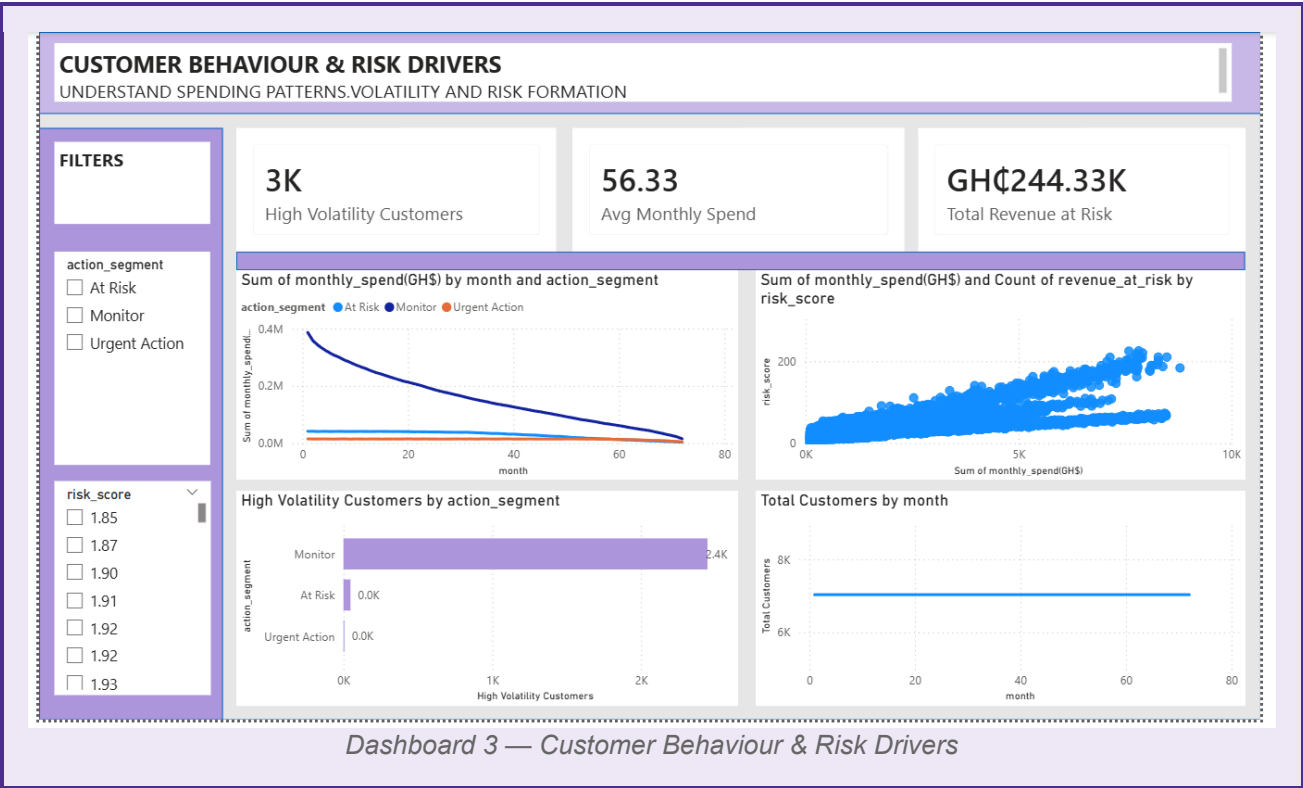


Figure 3: Behaviour Dashboard — spending volatility, risk score distribution, and segment patterns

Metric	Value	Significance
High Volatility Customers	2,442	Strong early churn predictor
Average Monthly Spend	GH¢56.38	Baseline spending across the customer base
Total Revenue at Risk	GH¢244,330	Consistent across all dashboard views
Most Volatile Segment	Monitor	High volume drives aggregate volatility

Dashboard 3 Insight

- 2,442 customers (~35% of the base) exhibit high spending volatility — a leading churn indicator preceding departure by 2-3 months.
- At Risk customers show the steepest declining spend trajectories despite lower volume.
- Higher monthly spend strongly correlates with higher risk score — the most valuable customers

need the most proactive attention.

4.4 Dashboard 4: Review Impact & Retention Strategy

The strategy dashboard translates risk intelligence into executable retention actions. It quantifies the revenue impact of each intervention tier and enables leadership to prioritise resources. This is the decision-making dashboard — built to change the room.

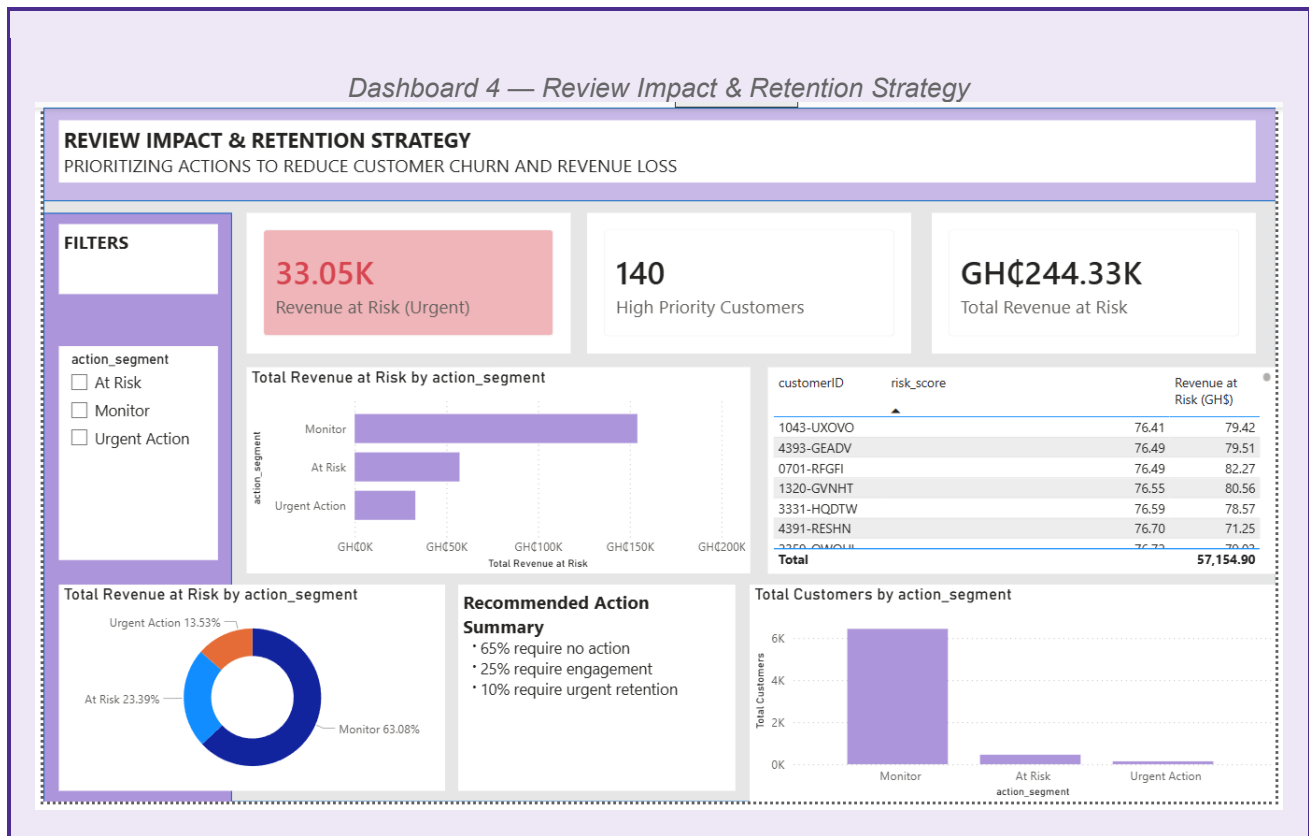


Figure 4: Retention Strategy Dashboard — revenue by action, customer priority list, and action distribution

Recommended Action	Customers	Revenue at Risk (GH¢)	% of Total
Monitor — No Action Required	~6,500	154,523.40	65.0%
Engagement Campaign / Usage Incentive	~400	57,154.90	25.1%
Call / Retain Offer / Discount Intervention	~140	33,050.00	9.9%
TOTAL	~7,000	244,728.30	100%

Dashboard 4 Insight

- The 140 Urgent Action customers control GH¢33,050 in revenue — each worth 10x a Monitor customer in risk intensity.
- The 400 At Risk customers contribute GH¢57,154 and respond well to structured engagement

campaigns.

- Monitor customers require sustained service quality and early warning monitoring to prevent migration upward.

5. Key Findings

5.1 Customer Segmentation

Segment	Count	% of Base	Revenue at Risk (GH¢)	Risk Intensity
Monitor (Low Risk)	~6,500	92.31%	154,523.40	Low per customer
At Risk (Moderate)	~400	5.70%	57,154.90	Moderate per customer
Urgent Action (High)	~140	1.41%	33,050.00	Highest per customer
Total	~7,000	100%	244,728.30	—

The distribution confirms a classic long-tail risk pattern: the majority of customers are stable, but the smallest segment carries the highest financial risk intensity per subscriber — a critical insight for resource allocation.

5.2 Revenue Exposure Analysis

Total revenue at risk across the customer base stands at GH¢244,330. This represents the estimated annual revenue exposure from customers exhibiting churn-predictive behaviour patterns.

Revenue Concentration Risk

- The Monitor segment holds 65% of total risk in absolute terms — driven entirely by volume.
- Each Urgent Action customer carries ~GH¢244 in revenue at risk vs ~GH¢25 per Monitor customer — a 10x difference.
- Retaining one Urgent customer is equivalent in value to stabilising ten Monitor customers.

5.3 Behavioural Risk Indicators

2,442 customers are identified as high volatility. Spending volatility is the strongest early predictor of churn identified in this analysis, preceding actual departure by an estimated two to three months — creating a viable intervention window.

6. Business Recommendations

6.1 Immediate: Urgent Retention (99 Customers)

- Assign dedicated relationship managers to the top 20 highest-risk customers
- Deploy personal phone calls within 72 hours of actioning this report
- Offer customised retention packages: loyalty discounts, service upgrades, plan restructuring
- Track weekly — further spend decline triggers escalated intervention

6.2 Short-Term: Engagement Campaigns (400 Customers)

- Launch targeted digital campaigns highlighting usage benefits and loyalty rewards
- Introduce bundled service packages that increase perceived value and switching costs
- Deploy automated re-engagement messages tied to declining spend triggers
- Schedule monthly check-in calls for the top 50 At Risk customers by revenue value

6.3 Ongoing: Monitoring Strategy (6,500 Customers)

- Maintain consistent service quality as the primary retention lever for this segment
- Implement automated early warning alerts when risk score increases by more than 10 points in a single month
- Conduct quarterly satisfaction surveys to detect dissatisfaction before behavioural decline

6.4 Systemic: Behavioural Tracking Infrastructure

- Automate the monthly refresh of risk scores using the Python model in this project
- Integrate Power BI dashboards into the weekly operations review for the commercial team
- Build a real-time spend volatility alert system flagging new high-volatility customers as they emerge
- Track retention campaign outcomes and use results to refine the risk model over time

7. Strategic Impact

If the retention framework recommended in this report is implemented, the projected outcomes across a 12-month horizon are as follows:

Impact Area	Current State	Target Outcome	Revenue Protected
Urgent customer churn	High — unmanaged	Reduce by 60% via direct intervention	GH¢19,830
At Risk campaign	No active programme	25% retention improvement via campaigns	GH¢14,288
Monitor early warning	No alert system	Prevent 5% migration into At Risk	GH¢7,975
Total Recoverable	—	—	GH¢40,000+

Strategic Value Statement

- ChurnGuard moves the telecom operator from reactive churn management to proactive retention intelligence.
- The Python risk model is scalable and can be run monthly on updated data with no additional infrastructure investment.
- The Power BI dashboards replace fragmented manual reporting and give every stakeholder a single source of truth.

8. Technical Appendix

8.1 Tools & Technologies

Tool / Technology	Purpose in This Project
Python (Pandas, NumPy)	Data cleaning, feature engineering, risk score calculation
SQL	Data validation, aggregation queries, segment verification
Power BI (DAX, Power Query)	Dashboard development, KPI cards, interactive filtering
Star Schema Modelling	Fact and dimension table design for the Power BI data model
IBM Telco Dataset	Source dataset — enhanced with simulated time-based spending

8.2 Risk Score Formula

- Decline frequency score: normalised count of months with negative spend change
- Revenue drop score: normalised magnitude of the largest single-month revenue decline
- Volatility score: standard deviation of monthly spend normalised to a 0-100 scale

8.3 Portfolio & Project Files

All Python notebooks, SQL queries, Power BI files, and the complete dataset are publicly available at:

github.com/emmanuella-danso

End of Report — ChurnGuard by Danso Emmanuella

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